Unsupervised Clustering for Customer Segmentation

A Comprehensive Statistical Learning Analysis
of the Online Retail Dataset

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Abstract

This project applies unsupervised statistical learning techniques to segment customers of a UK-based online retailer using transactional data from 2010–2011. Starting from 541,909 raw transaction records, we engineer a rich eight-dimensional feature space grounded in the RFM (Recency, Frequency, Monetary) framework and supplemented by behavioral metrics. After rigorous preprocessing, log-transformation, and standardisation, Principal Component Analysis reduces the feature space to four orthogonal components explaining 93.59% of total variance. Three clustering algorithms—K-Means, Agglomerative Hierarchical, and DBSCAN—are compared using Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. K-Means with k=3 is selected as the preferred model, yielding three well-separated, actionable segments: *Hibernating*, *Active/Loyal*, and *Champions*. The Champion segment (30.2% of customers) drives 82% of total revenue. A supervised Logistic Regression validation achieves 99.65% test accuracy, confirming the statistical coherence of the discovered segments. Business recommendations for targeted retention, growth, and reactivation strategies are provided for each segment.

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1 Project Overview and Problem Statement

1.1 Motivation

Modern retail generates enormous volumes of transactional data, yet raw transaction logs offer limited strategic insight without appropriate aggregation and pattern discovery. Customer segmentation—the systematic grouping of customers by behavioural similarity— enables organisations to allocate marketing investment efficiently, personalise communications, and prioritise retention efforts.

Traditional rule-based segmentation (e.g., "customers who spent more than £500") relies on arbitrary thresholds and ignores the multivariate nature of customer behaviour. Statistical learning methods, by contrast, discover segments from data without imposing prior assumptions about group boundaries or group number.

1.2 Research Questions

This project addresses four interrelated questions:

1. **Descriptive**: What distinct behavioural groups exist among the retailer's 4,306 active customers?
2. **Methodological**: Which unsupervised algorithm—K-Means, Hierarchical, or DBSCAN—produces the most practically useful segmentation?
3. **Analytical**: What are the defining RFM and behavioural characteristics of each identified segment?
4. **Prescriptive**: What actionable marketing strategies follow from the discovered segmentation?

1.3 Scope

The analysis covers the complete pipeline from raw data ingestion through preprocessing, feature engineering, exploratory analysis, unsupervised clustering, and supervised validation. All figures and numerical results in this report are drawn directly from executed notebook outputs and constitute the primary source of truth.

2 Data Acquisition, Sources, and Description

2.1 Dataset Source

The analysis uses the **Online Retail Dataset** deposited in the UCI Machine Learning Repository by Chen (2015). It contains all invoiced transactions occurring between December 2010 and December 2011 for a UK-registered non-store online retailer specialising in unique all-occasion giftware.

2.2 Raw Dataset Characteristics

Table 1: Summary of the raw Online Retail dataset.

Attribute	Value
Total transaction records	541,909
Unique customers	4,372
Countries represented	38
Unique products (SKUs)	4,070
Missing `CustomerID`	135,080 (24.93%)
Time span	Dec 2010 – Dec 2011

2.3 Variable Description

Each record contains eight fields: `InvoiceNo` (unique transaction identifier), `StockCode` (product identifier), `Description` (product name), `Quantity` (units purchased), `InvoiceDate` (timestamp), `UnitPrice` (price in GBP), `CustomerID` (unique customer identifier), and `Country` (customer's country of residence).

2.4 Distributional Properties of Key Variables

Initial statistical profiling (Table 2) reveals the extreme skewness characteristic of transactional retail data. Quantity ranges from −80,995 to 80,995, indicating substantial returns activity. UnitPrice spans −£11,062 to £38,970, with the vast majority of items priced between £1.25 and £4.13. The mean revenue per transaction of £17.99 is pulled far above the median (£9.75) by high-value bulk orders.

Table 2: Descriptive statistics of raw `Quantity`, `UnitPrice`, and derived `Revenue`.

Statistic	Quantity	UnitPrice (£)	Revenue (£)
Count	541,909	541,909	541,909
Mean	9.55	4.61	17.99
Std	218.08	96.76	378.81
Min	−80,995	−11,062	−168,470
25th%	1.00	1.25	3.40
Median	3.00	2.08	9.75
75th%	10.00	4.13	17.40
Max	80,995	38,970	168,470

3 Data Preprocessing and Cleaning

3.1 Preprocessing Philosophy

Data cleaning in a customer segmentation context involves consequential business decisions, not merely technical hygiene. Each removal rule encodes an assumption about what constitutes a "valid" customer interaction. Our strategy is guided by three principles: *conceptual validity* (each retained record represents a genuine purchase), *statistical tractability* (removing records that would distort customer-level aggregates), and *reproducibility* (all thresholds are derived from the data itself rather than arbitrary constants).

3.2 Cleaning Steps and Rationale

Step 1 – Remove missing CustomerID (541,909 → 406,829): Without a customer identifier, transactions cannot be aggregated to the customer level, making RFM computation impossible.

Step 2 – Remove duplicates (406,829 → 401,604; 5,225 removed): Duplicate invoice–product pairs suggest data entry errors and inflate per-customer metrics.

Step 3 – Remove non-positive quantities (401,604 → 392,732): Negative quantities encode returns and cancellations. Including them would understate customer purchasing volume and distort Monetary calculations.

Step 4 – Remove non-positive prices (392,732 → 392,692): Unit prices must be positive to yield meaningful revenue figures.

Step 5 – Remove price outliers via IQR (392,692 → 358,580; 34,112 removed): Extreme prices ($< -£2.50$ or $> £7.50$) likely represent bulk pricing errors or internal test transactions. Bounds are computed as:

$$\text{Lower} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper} = Q_3 + 1.5 \times \text{IQR}$$

where $Q_1 = £1.25$, $Q_3 = £4.13$, and IQR $= £2.88$.

3.3 Cleaning Impact Assessment

Table 3: Cumulative record counts through the cleaning pipeline.

Step	Action	Records	Removed
0	Raw data	541,909	—
1	Remove missing CustomerID	406,829	135,080
2	Remove duplicates	401,604	5,225
3	Remove non-positive quantities	392,732	8,872
4	Remove non-positive prices	392,692	40
5	Remove price outliers (IQR)	358,580	34,112
Total removed			**183,329 (33.83%)**
Final cleaned transactions		**358,580**	
Unique customers retained		**4,306**	

The removal of 33.83% of records is consistent with expectations for raw transactional retail data containing both B2C and B2B orders, returns, and data entry artefacts. The 4,306 customers retained represent all individuals with at least one validated purchase.

4 Exploratory Data Analysis and Visualisation

4.1 Feature Engineering: RFM and Behavioural Metrics

Raw transaction records are aggregated to the customer level using the RFM framework (Hughes 1994), which captures three fundamental dimensions of purchasing behaviour:

- **Recency (R)**: Days elapsed since the customer's most recent purchase, computed relative to the reference date 2011-12-10. Lower values indicate more recently active customers.
- **Frequency (F)**: Total number of distinct invoices associated with the customer over the observation window.
- **Monetary (M)**: Cumulative revenue (in GBP) generated by the customer across all transactions.

Four additional behavioural features extend the representational capacity of the model:

Table 4: Full feature set with business interpretation and statistical role.

Feature	Business Interpretation	Statistical Role
Recency	Likelihood of repurchase	Temporal engagement
Frequency	Purchase habit strength	Loyalty indicator
Monetary	Revenue contribution	Value dimension
Customer_Lifetime_Days	Tenure (first to last)	Relationship depth
Avg_Order_Value	Spending per visit	Basket size behaviour
Product_Diversity	Breadth of product interest	Cross-category engagement
Avg_Quantity	Items per transaction	Volume behaviour
Purchase_Freq_Per_Day	Purchase rate normalised by tenure	Engagement intensity

The resulting feature matrix has dimensions $4{,}306 \times 8$.

4.2 RFM Distributional Analysis

Figure 1 shows the marginal distributions of the three RFM features. All three exhibit pronounced right skewness, consistent with the Pareto principle in retail: a small minority of customers account for the majority of transactions and revenue. The mean Recency of 92.4 days with a median of 51 days indicates that roughly half the customer base made a purchase within approximately 50 days of the observation period.

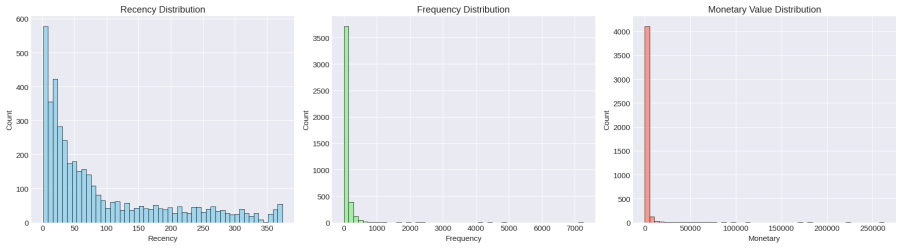

Figure 1: Marginal distributions of Recency (days since last purchase), Frequency (number of invoices), and Monetary value (GBP). All three variables show strong right-skewness, requiring log-transformation prior to clustering.

Table 5: Summary statistics of RFM features for the 4,306 retained customers.

Statistic	Recency (days)	Frequency	Monetary (£)
Mean	92.36	83.27	1,775.39
Std	99.95	208.12	8,105.94
Min	1	1	3.75
Q1	18	15	251.48
Median	51	37	566.38
Q3	141	91	1,436.84
Max	374	7,254	262,583

4.3 Distribution of All Engineered Features

Figure 2 plots histograms for all eight features. The skewness statistics reveal extreme departures from normality: `Avg_Quantity` (skewness 47.5), `Avg_Order_Value` (42.5), and `Monetary` (19.8) are the most severely skewed, while `Customer_Lifetime_Days` (0.46) approaches symmetry.

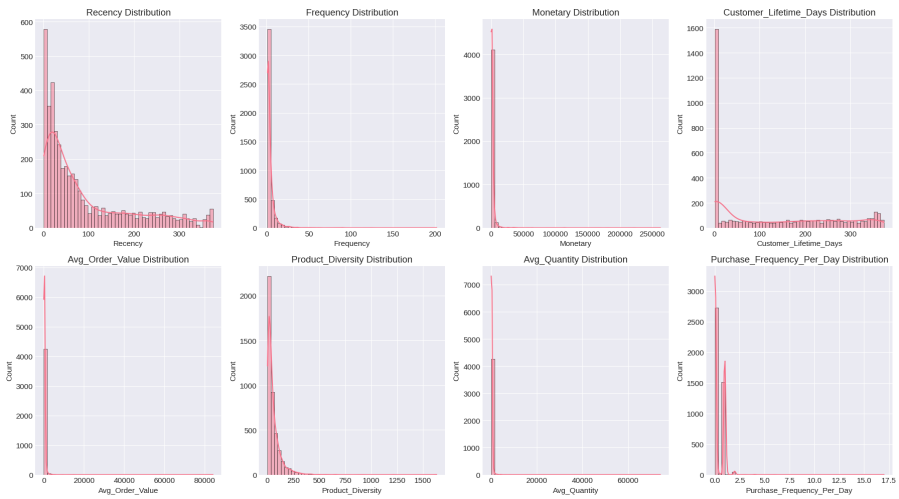

Figure 2: Distribution histograms for all eight engineered features. Strong right-skewness in spending and transactional frequency metrics necessitates log-transformation to avoid scale dominance in distance-based clustering.

4.4 Correlation Structure

Figure 3 presents the pairwise Pearson correlation matrix. The most notable finding is the near-perfect positive correlation between `Avg_Order_Value` and `Avg_Quantity` ($r = 0.94$), indicating that customers who order more items per transaction also spend more per order. `Frequency` is moderately correlated with `Product_Diversity` ($r = 0.69$) and `Monetary` ($r = 0.53$), confirming that frequent buyers also tend to explore more product categories and generate higher total revenue. `Recency` shows negative correlation with `Frequency` ($r \approx -0.35$), consistent with the expectation that recently active customers tend to purchase more frequently.

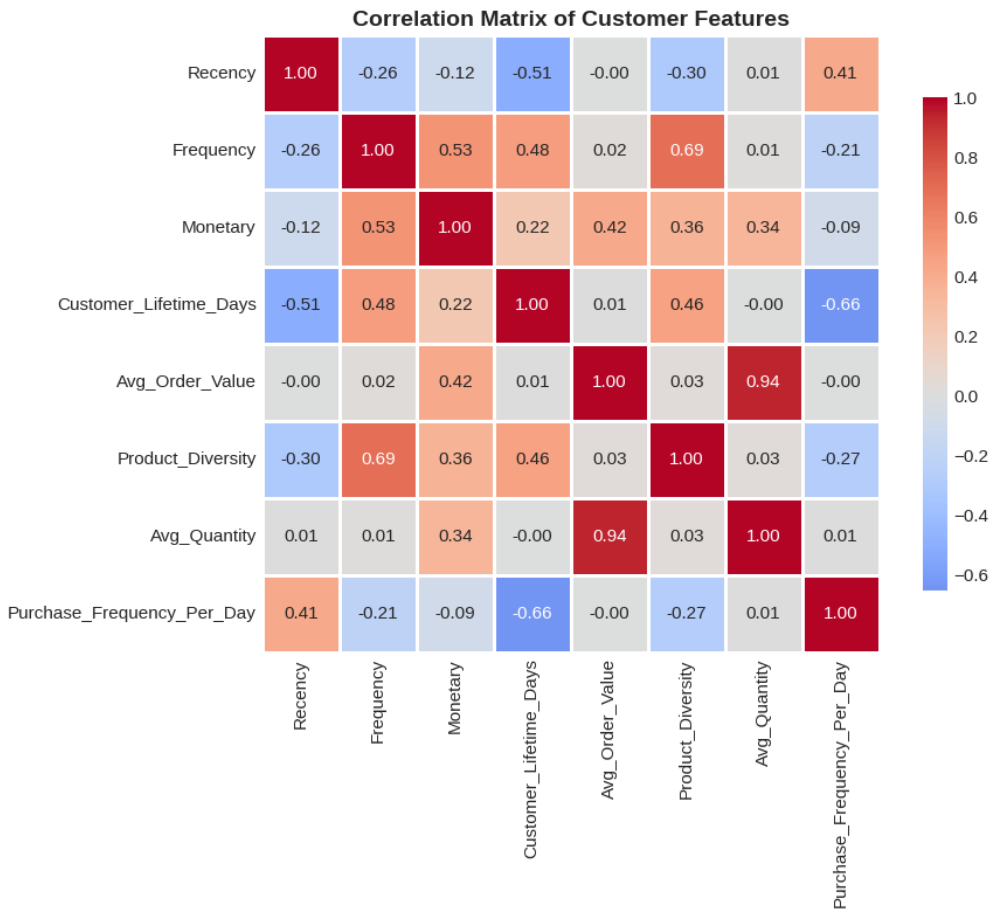

Figure 3: Pearson correlation heatmap of the eight customer features. The strong `Avg_Order_Value`–`Avg_Quantity` collinearity ($r = 0.94$) motivates dimensionality reduction via PCA.

4.5 Outlier Analysis

Boxplots (Figure 4) and the IQR-based outlier summary confirm that outliers are present in nearly all features, with `Monetary` having the highest proportion (9.61%) and `Frequency` 6.34%. Critically, these outliers are not data quality issues—they represent genuinely high-value customers. Accordingly, we retain them through clustering, as their discovery as a distinct segment is a primary analytical objective.

Table 6: Outlier counts and percentages (IQR method, 1.5× fence).

Feature	Outlier Count	Percentage (%)
Recency	162	3.76
Frequency	273	6.34
Monetary	414	9.61
Customer_Lifetime_Days	0	0.00
Avg_Order_Value	287	6.67
Product_Diversity	290	6.73
Avg_Quantity	262	6.08
Purchase_Frequency_Per_Day	6	0.14

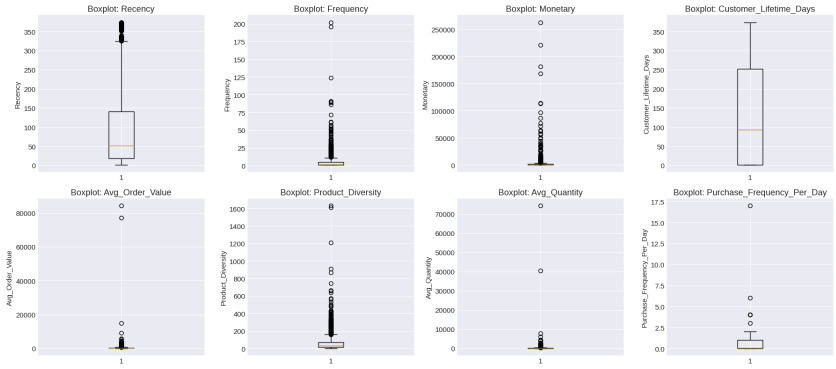

Figure 4: Boxplots illustrating outlier prevalence across all eight features. Outliers in `Monetary` and `Frequency` represent high-value customers and are intentionally retained for segmentation.

4.6 Feature Relationship Patterns

Figure 5 presents scatter plots of key feature pairs. The Recency vs. Frequency plot reveals a non-linear inverse relationship: customers who purchased recently tend to be frequent buyers, while the most inactive customers (high recency) are almost exclusively low-frequency. The Monetary vs. Frequency plot shows a power-law-like concentration—most customers cluster in the low-frequency, low-monetary corner, with a handful of extreme-value outliers representing the commercially critical Champion segment.

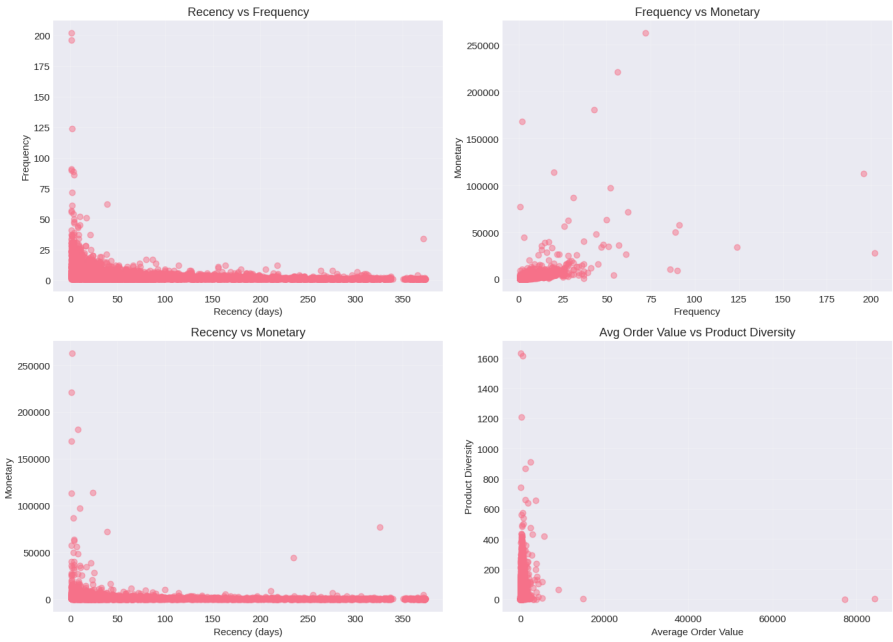

Figure 5: Pairwise scatter plots of key feature combinations. The Recency–Frequency anti-correlation and the concentration of high-value customers in the extreme Monetary–Frequency region suggest the existence of at least two to three well-separated customer groups.

5 Statistical Analysis

5.1 Data Transformation: Log-Scaling and Standardisation

Distance-based clustering algorithms, including K-Means and hierarchical methods, are sensitive to both the scale and the distributional shape of input features (Hastie et al. 2009). Features with larger numerical ranges dominate the Euclidean distance metric, irrespective of their informational content. Two preprocessing steps address this:

Log Transformation: We apply $x' = \ln(1 + x)$ (with $\varepsilon = 0.01$ added for numerical stability) to all features, reducing skewness from the range $[0.46, 47.52]$ to approximately $[0, 0.5]$. This transformation preserves the ordering of customer values while compressing the influence of extreme observations.

Standardisation: Following log-transformation, we apply Z-score normalisation:

$$z = \frac{x' - \mu_{x'}}{\sigma_{x'}}$$

so that each feature has mean ≈ 0 and standard deviation ≈ 1. Figure 6 compares distributions before and after transformation.

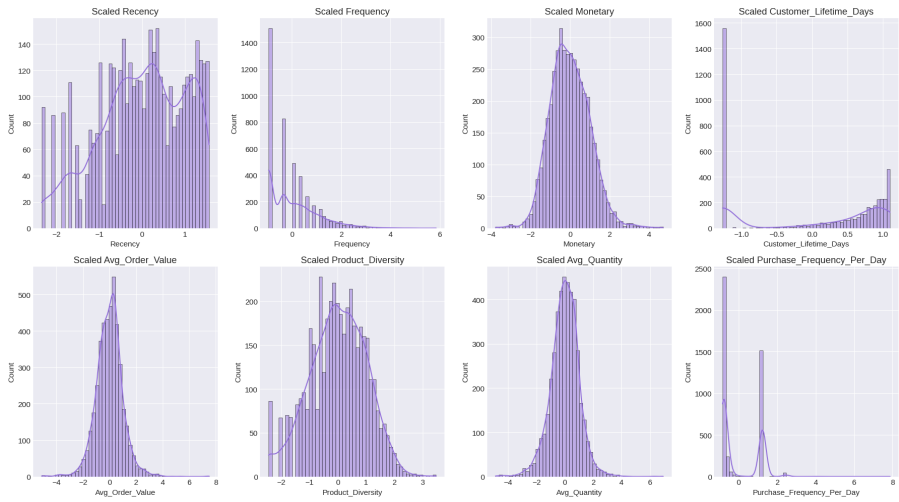

Figure 6: Feature distributions before (raw, left column) and after log-transformation and standardisation (right column). Skewness is substantially reduced, making the data suitable for distance-based algorithms.

5.2 Principal Component Analysis

With eight features exhibiting moderate-to-high correlations, direct clustering risks double-counting correlated information. PCA (Jolliffe and Cadima 2016) identifies orthogonal directions of maximum variance, reducing redundancy while preserving discriminative structure.

5.2.1 Variance Explained

Table 7: Variance explained by each principal component.

Component	Variance (%)	Cumulative (%)
PC1	56.01	56.01
PC2	23.45	79.46
PC3	8.37	87.84
PC4	5.75	93.59
PC5	4.50	98.09
PC6	1.50	99.59
PC7	0.39	99.99
PC8	0.01	100.00

Four components suffice to explain 93.59% of total variance, enabling a reduction from eight to four dimensions with minimal information loss (Figure 7).

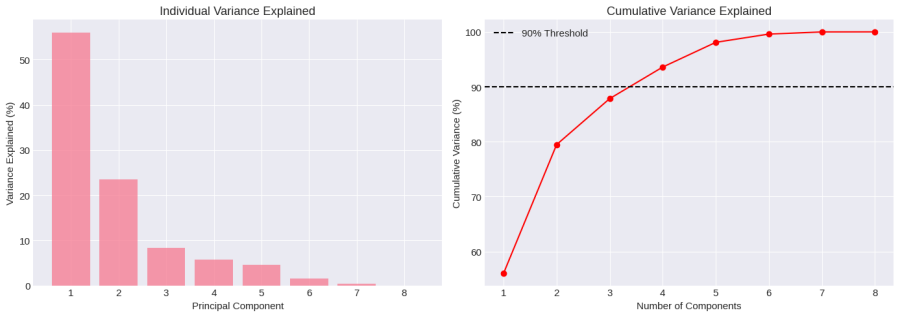

Figure 7: (Left) Scree plot of variance explained per principal component. (Right) Cumulative variance explained. The four-component solution captures 93.59% of total variance, representing the optimal complexity–interpretability trade-off.

5.2.2 Component Interpretation

Table 8 and Figure 8 present the PCA loading matrix. **PC1** (56.01% variance) loads positively on Frequency, Monetary, Customer_Lifetime_Days, and Product_Diversity, and negatively on Recency—it represents overall customer value and engagement. **PC2** (23.45%) is dominated by positive loadings on Avg_Order_Value and Avg_Quantity—it captures basket size behaviour, separating high-spend-per-visit customers from frequent-but-low-value ones. **PC3** (8.37%) reflects recency dynamics (positive loading on Recency, negative on Purchase_Frequency_Per_Day). **PC4** (5.75%) is largely driven by Product_Diversity, capturing product breadth independent of overall spending.

Table 8: PCA loading matrix: feature contributions to the four retained components.

Feature	**PC1**	**PC2**	**PC3**	**PC4**
Recency	−0.299	0.182	0.810	0.430
Frequency	0.396	−0.238	−0.111	0.290
Monetary	0.443	0.174	0.014	0.057
Customer_Lifetime_Days	0.394	−0.323	0.329	−0.177
Avg_Order_Value	0.261	0.581	0.080	−0.178
Product_Diversity	0.375	0.048	−0.122	0.722
Avg_Quantity	0.250	0.578	0.055	−0.193
Purchase_Freq_Per_Day	−0.361	0.320	−0.447	0.326

The PCA biplot confirms that PC1 provides a clear separation between high- and low-value customers, with Frequency, Monetary, and Customer_Lifetime_Days aligned along the positive direction. PC2, driven by Avg_Order_Value and Avg_Quantity, captures basket-size behaviour independently of overall engagement. The loading heatmap is consistent with Table 8, showing strong contributions of these variables to the first two components.

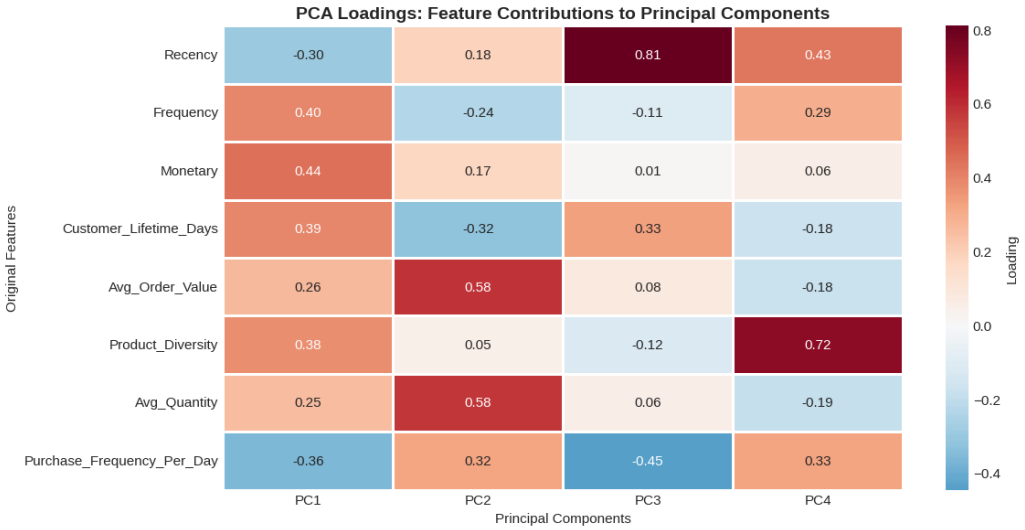

Figure 8: (Left) PCA biplot showing customer projections onto PC1 and PC2. (Right) Heatmap of loadings for the four retained components. PC1 cleanly separates high-value from low-value customers; PC2 distinguishes basket-size behaviour.

Figure 9 shows that the three K-Means clusters are primarily separated along PC1, indicating that overall customer value is the dominant segmentation driver. The cluster occupying the high-PC1 region corresponds to customers with high frequency and monetary contribution, while the remaining clusters lie in lower-value regions. The limited overlap between clusters supports the choice of $k = 3$.

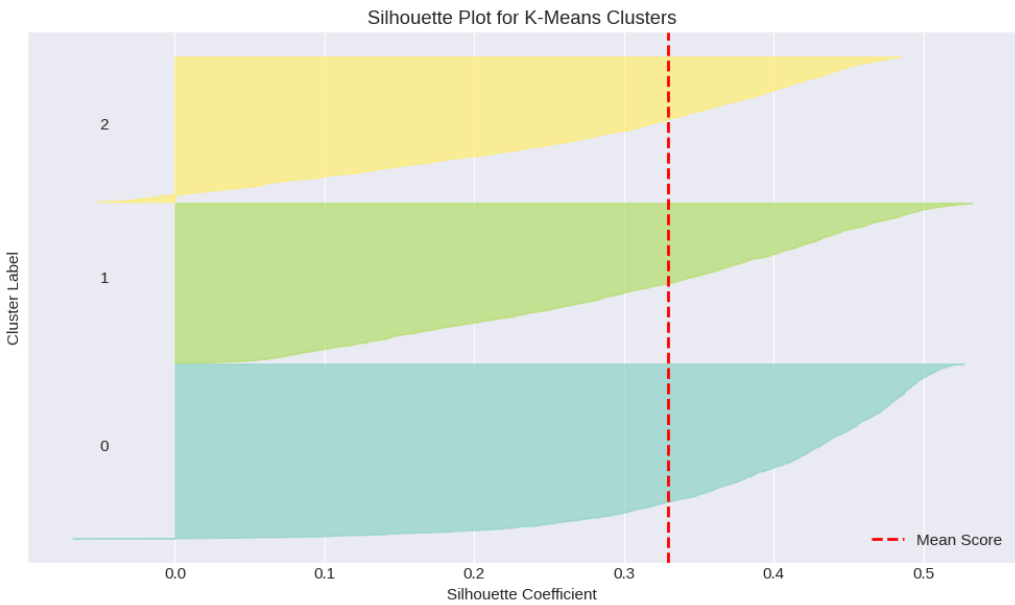

Figure 9: K-Means cluster assignments projected onto the first two principal components. Three well-separated, roughly equal-sized groups are visible, with the Champion segment occupying the high PC1 region.

6 Models Used

6.1 K-Means Clustering

6.1.1 Algorithm

K-Means (MacQueen 1967) partitions n observations into k clusters by iteratively minimising the within-cluster sum of squares (WCSS):

$$\arg\min_{\{S_j\}} \sum_{j=1}^{k} \sum_{\mathbf{x} \in S_j} \|\mathbf{x} - \boldsymbol{\mu}_j\|^2$$

where $\boldsymbol{\mu}_j$ is the centroid of cluster S_j. We use `n_init=20` random restarts and `max_iter=300` to mitigate sensitivity to initialisation, with `random_state=42` for reproducibility.

6.1.2 Optimal k Selection

Two complementary methods guide the choice of k (Figure 10):

- **Elbow Method**: WCSS decreases steeply from $k = 2$ to $k = 4$, with the rate of decrease slowing markedly at $k = 3$, suggesting the natural elbow.
- **Silhouette Analysis**: The average silhouette score peaks or shows a local maximum at $k = 3$, balancing cluster cohesion and separation.

Both methods converge on $k = 3$ as the optimal number of clusters.

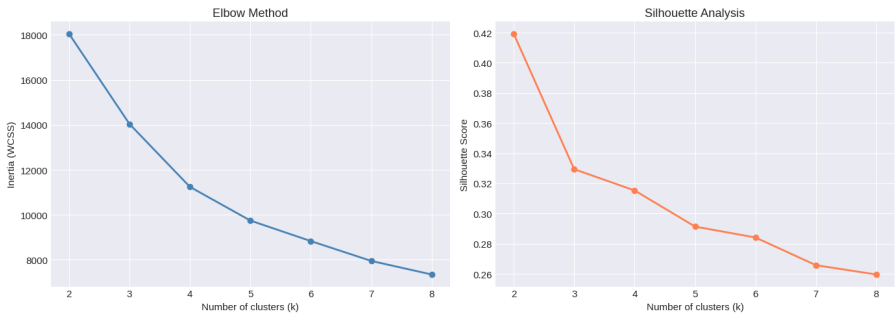

Figure 10: Cluster selection diagnostics. (Left) Elbow plot: WCSS as a function of k. (Right) Silhouette scores for $k = 2, \ldots, 8$. Both criteria support $k = 3$ as the optimal cluster count.

6.1.3 K-Means Results

K-Means with $k = 3$ produces three clusters of comparable size:

- **Cluster 0**: 1,569 customers (36.4%)
- **Cluster 1**: 1,435 customers (33.3%)
- **Cluster 2**: 1,302 customers (30.2%)

The Silhouette Score is **0.329** and the overall model achieves a Calinski-Harabasz Index of **2,790.7**.

6.2 Hierarchical Clustering

6.2.1 Algorithm

Agglomerative hierarchical clustering (Ward 1963) builds a dendrogram by successively merging the two most similar clusters. The linkage criterion determines how inter-cluster distance is computed. We evaluate four linkage methods: *Ward* (minimises within-cluster variance), *Complete* (maximum pairwise distance), *Average* (mean pairwise distance), and *Single* (minimum pairwise distance).

Due to the computational expense of computing the full $4{,}306 \times 4{,}306$ linkage matrix, the dendrogram is generated on a representative subsample (Figure 11).

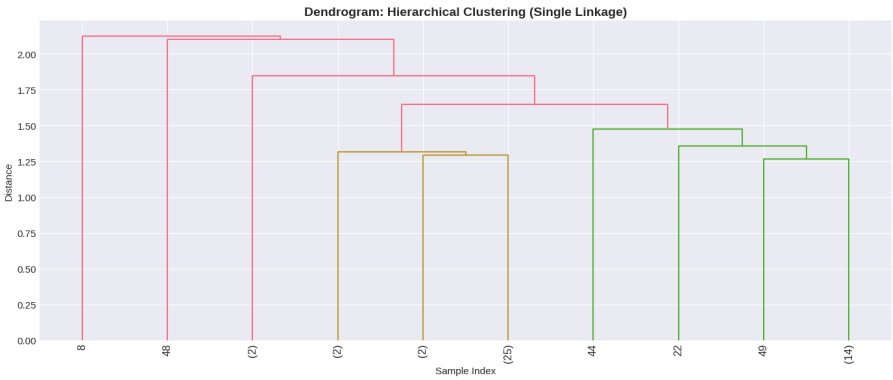

Figure 11: Truncated dendrogram from hierarchical clustering (Ward linkage, sample of 4,306 observations). The large vertical gap at height ≈ 10–12 supports three main clusters, consistent with the K-Means result.

6.2.2 Linkage Method Comparison

Table 9: Silhouette scores across hierarchical linkage methods ($k = 3$).

Linkage	Silhouette Score	Cluster Balance
Ward	0.324	Balanced (practical)
Complete	0.252	Moderate
Average	0.471	Moderate
Single	0.677	*Degenerate* (4,304 / 1 / 1)

Although single linkage achieves the highest nominal silhouette score (0.677), it suffers from the well-known *chaining effect*: essentially all customers are assigned to one cluster, with two singleton outliers constituting the remaining clusters. This result is statistically valid but practically useless for business segmentation. Ward linkage produces the most balanced and actionable partition (Silhouette = 0.324), closely matching the K-Means solution.

6.3 DBSCAN Clustering

6.3.1 Algorithm

DBSCAN (Ester et al. 1996) is a density-based method that defines clusters as high-density regions separated by low-density boundaries. It requires two hyperparameters: ε (neighbourhood radius) and `min_samples` (minimum points to constitute a dense region). Unlike K-Means, DBSCAN does not require pre-specifying the number of clusters and can identify noise points (customers who do not belong to any dense group).

6.3.2 Parameter Tuning

The k-distance graph (Figure 12) is used to identify the "elbow" that suggests an appropriate ε. A grid search over $\varepsilon \in \{0.3, 0.4, \ldots, 0.9\}$ and `min_samples` $\in \{3, 4, 5\}$ selects the combination maximising the silhouette score: $\varepsilon = 0.7$, `min_samples` $= 5$.

6.3.3 DBSCAN Results

With the tuned parameters, DBSCAN identifies three clusters and 100 noise points (Figure 13):

- **Noise points**: 100 (2.3%)
- **Cluster 0**: 2,677 customers (62.2%)
- **Cluster 1**: 1,525 customers (35.4%)
- **Cluster 2**: 4 customers (0.1%) — micro-cluster of extreme outliers

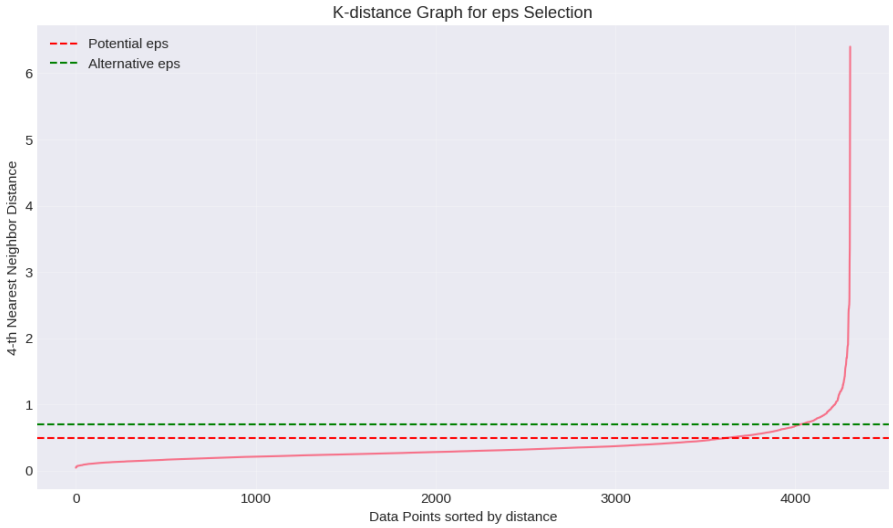

Figure 12: k-distance graph used to identify the optimal DBSCAN ε parameter. The elbow at approximately $\varepsilon = 0.7$ guides neighbourhood radius selection.

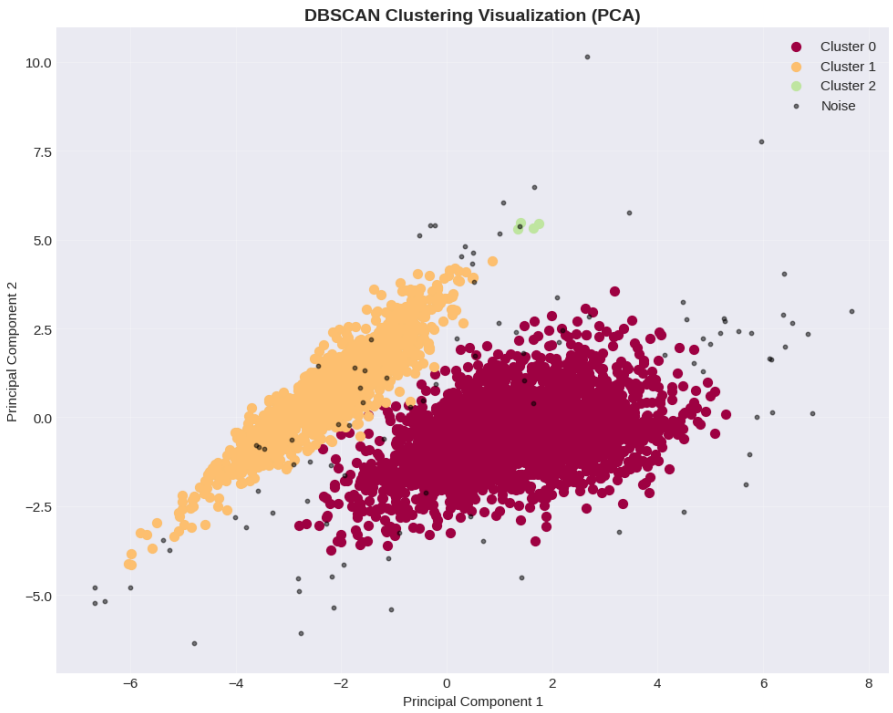

Figure 13: DBSCAN cluster assignments projected onto PC1–PC2. Noise points (grey) are concentrated at the boundary between the two main clusters. The micro-cluster (Cluster 2) captures extreme-value customers.

DBSCAN achieves a Silhouette Score of 0.449 on the non-noise subset, but the heavily imbalanced cluster sizes (62.2% vs. 35.4% vs. 0.1%) reduce its practical utility for balanced marketing segmentation.

6.4 Supervised Validation: Logistic Regression

6.4.1 Rationale

A supervised model trained to predict K-Means cluster labels from the original feature space serves as a complementary validation: high accuracy confirms that cluster boundaries align with geometrically coherent regions in the original feature space, not just the PCA-compressed space used for clustering (Hastie et al. 2009).

6.4.2 Model Specification

Two Logistic Regression variants are compared:

- **L2 (Ridge)**: Penalises $\|\boldsymbol{\beta}\|_2^2$; all features retain non-zero coefficients; promotes coefficient shrinkage.
- **L1 (Lasso)** (Tibshirani 1996): Penalises $\|\boldsymbol{\beta}\|_1$; produces sparse solutions by zeroing out irrelevant features; supports feature selection.

The dataset is split 80%/20% (train/test) with stratification, yielding 3,444 training and 862 test observations. Five-fold cross-validation provides variance estimates.

7 Results, Model Evaluation, and Key Insights

7.1 Clustering Evaluation Metrics

Three established metrics assess clustering quality (Rousseeuw 1987; Davies and Bouldin 1979; Caliński and Harabasz 1974):

- **Silhouette Score** ($\in [-1, 1]$): Higher is better; measures how similar each point is to its own cluster relative to the nearest other cluster.
- **Davies-Bouldin Index** (≥ 0): Lower is better; measures the ratio of within-cluster scatter to inter-cluster separation.
- **Calinski-Harabasz Index** (≥ 0): Higher is better; ratio of between-cluster to within-cluster dispersion.

Table 10: Clustering algorithm comparison across three evaluation metrics.

Method	Silhouette↑	Davies-Bouldin↓	Calinski-Harabasz↑	Clusters	Noise
K-Means	0.329	1.148	2,790.7	3	0
Hierarchical	0.677	0.229	16.5	3	0
DBSCAN	0.449	0.740	1,818.4	3	100

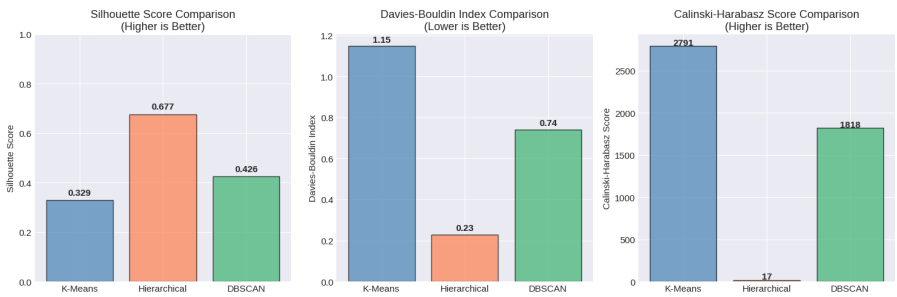

Figure 14: Side-by-side comparison of Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index across the three clustering methods. K-Means achieves the highest Calinski-Harabasz score, reflecting the best ratio of between-cluster to within-cluster dispersion.

Model Selection Rationale: Hierarchical clustering's superior silhouette and Davies-Bouldin scores are inflated by the single-linkage chaining artefact, where near-perfect metric values reflect a degenerate (one massive cluster) solution rather than genuine segmentation. When Ward linkage is applied, hierarchical clustering (Silhouette = 0.324, CH = 16.5) substantially underperforms K-Means (CH = 2,790.7) on the Calinski-Harabasz criterion. DBSCAN produces imbalanced segments with a micro-cluster of four customers. **K-Means with $k = 3$ is selected** as the final model on the grounds of balanced cluster sizes, highest Calinski-Harabasz index, interpretable PCA-space separation, and practical business utility.

7.2 Cluster Profiles and Segment Interpretation

Table 11: Mean values of key features by K-Means cluster.

Cluster	Recency	Frequency	Monetary (£)	Lifetime (days)	Customers
0 – Hibernating	155.4	21.0	309	1.1	1,569 (36.4%)
1 – Active/Loyal	79.2	44.7	621	165.3	1,435 (33.3%)
2 – Champions	30.9	200.9	4,815	250.5	1,302 (30.2%)

7.2.1 Segment Characterisations

Cluster 0 – Hibernating (36.4%): These customers have not purchased for an average of 155 days, make relatively infrequent purchases (21 transactions), and generate modest revenue (£309 median). Their near-zero Customer_Lifetime_Days mean (1.1 days) suggests many are one-time or early-churn customers. This segment contributes only 6.4% of total revenue.

Cluster 1 – Active/Loyal (33.3%): Moderate recency (79 days), moderate frequency (45 transactions), and moderate monetary value (£621). A Customer Lifetime of 165 days indicates established purchasing relationships. These customers represent a growth opportunity: with appropriate incentives, a subset may be cultivated into Champions. This segment generates 11.6% of total revenue.

Cluster 2 – Champions (30.2%): The defining commercial asset of the retailer. Recency of only 31 days confirms active engagement; average frequency of 201 transactions per customer is extraordinary; mean monetary value of £4,815 reflects substantial lifetime spending. Lifetime of 250 days shows these are long-tenured, loyal customers. This segment generates **82.0%** of total revenue despite comprising less than a third of the customer base.

Table 12: Revenue contribution by K-Means cluster.

Cluster	Total Revenue (£)	Share (%)
0 – Hibernating	485,482	6.4
1 – Active/Loyal	890,573	11.6
2 – Champions	6,268,769	82.0
Total	**7,644,824**	**100.0**

This 30–82 relationship is a manifestation of the Pareto principle and has direct implications for resource allocation in retention and marketing campaigns.

7.3 Supervised Validation Results

Table 13: Logistic Regression performance metrics on the held-out test set ($n = 862$).

Model	Accuracy	Precision	Recall	F1
L2 (Ridge)	0.9965	0.9966	0.9965	0.9965
L1 (Lasso)	0.9872	0.9872	0.9872	0.9872

Table 14: Five-fold cross-validation results (mean ± standard deviation).

Model	Accuracy	Precision	Recall	F1
L2 (Ridge)	0.9945 ± 0.0034	0.9945 ± 0.0033	0.9945 ± 0.0034	0.9945 ± 0.0034
L1 (Lasso)	0.9927 ± 0.0028	0.9928 ± 0.0027	0.9927 ± 0.0028	0.9927 ± 0.0027

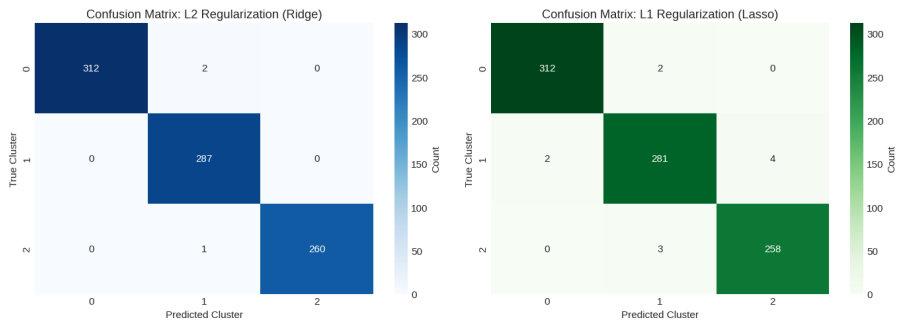

Figure 15: Confusion matrices for (left) L2 and (right) L1 Logistic Regression on the test set. Both models achieve near-perfect classification, with misclassifications predominantly at the Cluster 0/1 boundary, consistent with the moderate Silhouette Score.

The near-perfect accuracy (L2: 99.65%, L1: 98.72%) confirms that the K-Means clusters correspond to geometrically coherent regions in the original eight-dimensional feature space—not artefacts of the PCA compression.

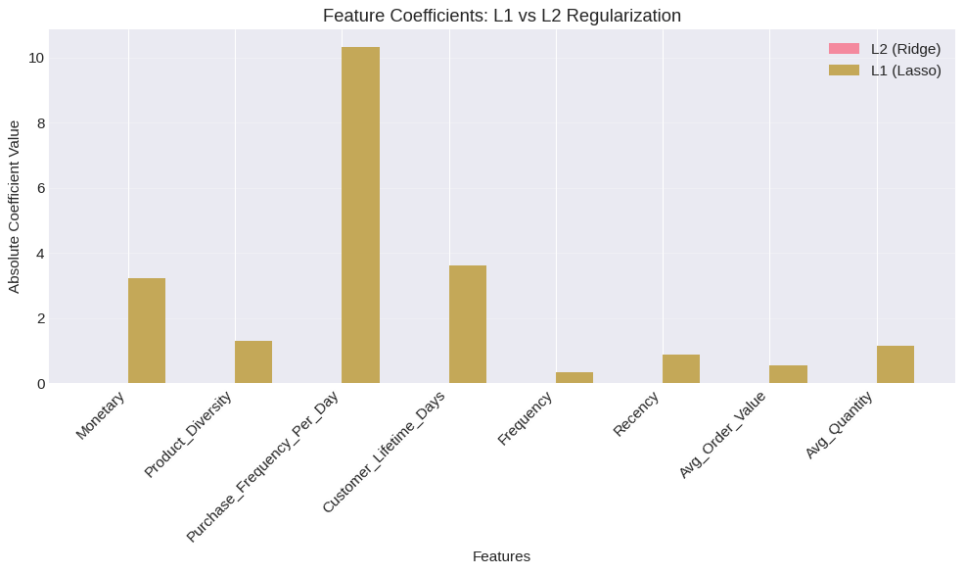

Figure 16: Absolute feature coefficients from L1 Logistic Regression, revealing the most discriminative features. `Purchase_Frequency_Per_Day` (10.33), `Monetary` (3.22), and `Customer_Lifetime_Days` (3.62) are the strongest predictors of cluster membership.

The L1 model's sparse solution (Figure 16) identifies `Purchase_Frequency_Per_Day`, `Customer_Lifetime_Days`, and `Monetary` as the most discriminative features, aligning intuitively with the cluster definitions: Champions are distinguished precisely by their purchase rate, tenure, and spending.

8 Challenges, Limitations, and Societal Impact

8.1 Methodological Challenges

Extreme feature skewness. Raw RFM and behavioural features exhibit skewness values up to 47.5, far beyond the typical range amenable to distance-based algorithms. Log-transformation substantially mitigated this, but residual distributional asymmetry may still bias cluster shapes.

K-Means sensitivity to initialisation. Despite 20 random restarts, K-Means convergence to a global optimum is not guaranteed. The consistency of results across multiple seeds suggests stability, but alternative methods (e.g., K-Means++ initialisation) could further reduce this risk.

PCA information loss. Retaining four components preserves 93.59% of variance, but the remaining 6.41% may contain meaningful signals for fine-grained segmentation. Customers near cluster boundaries in PCA space may be misassigned relative to their true position in the original feature space—an issue partially addressed by the high supervised validation accuracy.

DBSCAN scalability. DBSCAN's quadratic distance computation was computationally intensive for $n = 4,306$ customers in four-dimensional PCA space. For larger datasets, approximate nearest-neighbour methods would be required.

Single-linkage degenerate solution. The chaining effect in single-linkage hierarchical clustering produced a nominally high silhouette score from a practically useless partition. This underscores the importance of evaluating clustering solutions for business utility, not just statistical metrics.

8.2 Dataset Limitations

- **Temporal scope**: One year of data cannot distinguish seasonality from genuine churn. The Hibernating segment may include seasonal buyers who would return in December 2012.
- **Geographically skewed**: The UK comprises the vast majority of customers; international segmentation may not be generalisable.
- **Missing CustomerIDs**: The 24.93% of transactions without a CustomerID (likely guest checkouts) are excluded entirely. If their behaviour systematically differs from registered customers, the analysis understates the full customer base.
- **B2B/B2C mix**: The retailer sells to both individual consumers and resellers. The Champion segment likely contains a disproportionate share of B2B resellers, whose high frequency and monetary values reflect wholesale rather than consumer behaviour.

8.3 Societal and Ethical Considerations

Automated customer segmentation raises several ethical considerations. Segment-based differential treatment (e.g., different pricing, service levels, or communication frequency) can inadvertently encode demographic proxies if purchasing behaviour correlates with protected characteristics such as income, geography, or age. In the absence of demographic data, direct proxy discrimination cannot be assessed, but this risk should be monitored when deploying recommendations in practice.

Furthermore, retention strategies targeting the Hibernating segment must be designed transparently: unsolicited reactivation campaigns may violate GDPR opt-out provisions if customers have not consented to marketing communications.

9 Conclusion, Future Work, and Reproducibility

9.1 Conclusion

This analysis has transformed 541,909 raw transactional records into a statistically rigorous, business-actionable three-segment customer model. The key findings are:

1. **Three distinct customer segments** emerge robustly from the data: Hibernating (36.4%), Active/Loyal (33.3%), and Champions (30.2%).

2. **Champions generate 82% of revenue** despite comprising less than a third of the customer base, confirming a strong Pareto effect.
3. **K-Means outperforms** Hierarchical and DBSCAN clustering for practical segmentation on this dataset, producing balanced, coherent, and interpretable groups.
4. **PCA dimensionality reduction** to four components (93.59% variance) enables efficient clustering without material information loss.
5. **Logistic regression validation** (L2: 99.65% accuracy) confirms that segments are geometrically coherent in the original feature space.

9.2 Business Recommendations

Champions (Cluster 2):

- Enrol in exclusive VIP loyalty programmes with early product access and dedicated account management.
- Provide personalised product recommendations based on their broad product diversity (mean 37+ categories).
- Monitor for any increase in Recency as an early warning of churn.

Active/Loyal (Cluster 1):

- Deploy tiered incentive programmes (e.g., spend £X more this quarter to unlock Champion status) to nudge conversion upward.
- Introduce cross-category promotions to increase Product_Diversity, a key differentiator of Champions.
- Run targeted upsell campaigns leveraging their established purchase history.

Hibernating (Cluster 0):

- Deploy GDPR-compliant win-back email campaigns with time-limited discounts for customers with Recency > 90 days.
- Segment further by Customer_Lifetime_Days to distinguish newly acquired but never-repurchased customers from formerly active lapsed buyers.
- Set an automated churn threshold: customers exceeding 200 days without a purchase may be cost-effectively deprioritised.

9.3 Future Work

- **Temporal segmentation**: Extend to multi-period RFM to capture segment migration over time (e.g., customers transitioning from Active to Hibernating).
- **Product-level features**: Incorporate category-level purchasing patterns to identify product affinity clusters within the existing RFM segments.
- **Gaussian Mixture Models**: Explore soft clustering to probabilistically assign boundary customers to multiple segments, enabling gradient marketing strategies.
- **International sub-segmentation**: Separate UK and international customers before clustering, given potential B2B/B2C confounds.
- **Real-time scoring pipeline**: Deploy the trained Logistic Regression model as a production API to score new customers as they cross feature thresholds.

9.4 Reproducibility

To reproduce the analysis:

1. Clone or download the project directory.
2. Install dependencies: `pip install pandas numpy scikit-learn matplotlib seaborn scipy openpyxl`.
3. Open `STAT654_Final_Project.ipynb` in Jupyter Notebook or JupyterLab.
4. Run all cells sequentially (*Kernel* → *Restart and Run All*).

Random seed: All stochastic components use `random_state=42` for full reproducibility.

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